1) Backprop by hand

MLP w/ 3 input nodes, 2 hidden layers, 3 outputs

$\downarrow$ ReLU activation $\qquad$ $\downarrow$ softmax

$$w^1 = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & -2 \end{bmatrix} \qquad b^1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$w^2 = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix} \qquad b^2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$w^3 = \begin{bmatrix} 2 & 2 \\ 3 & -3 \\ 2 & 1 \end{bmatrix} \qquad b^3 = \begin{bmatrix} 0 \\ -4 \\ -2 \end{bmatrix}$$

a) feedforward computation.

input $\rightarrow x = [+1 \ -1 \ +1]^T$

layer 1 : ReLU

$$s^1 = w^1 x + b^1$$

$$s(1)_1 = (1)(1) + (-2)(-1) + (1)(1) + 1$$

$$= 1 + 2 + 1 + 1 = 5$$

$$s(1)_2 = 3(1) + (4)(-1) + (1)(-2) + -2$$

$$= 3 - 4 - 2 - 2 = -5$$

$$s(1) = [5, -5]^T$$

$$a(1) = \text{ReLU}(s1) = \max(0, s1)$$

$a(1) = [5, 0]^T$

$\dot{a}(1) = \text{Relu}'\ s(1) = 1\{s>0\}$

$\dot{a}(1) = [1, 0]^T$ 1 → $s > 0$

0 when $s \leq 0$.

layer 2.

$s(2) = W_2 a_1 + b_2$

$s_2\ 2 = (3)(5) + 4(0) + 0 = 15$

$s(2) = [6, 15]^T$

$a(2) = (1, 1)^T$

layer 3 softmax.

$s(3)\ (w(3)\ (a2^T) + b(3))$

$s(3)(1 = 2(6) + 2(15) + 0$

$= 42$

$s(3)\quad 6(3) + -3(15) + -4$

$=$

$-$ -31

$$S(3) = 2(6) + 1(15) + -2$$
$$= 25$$

$$S(3) = [42, -31, 25]^T$$

$$a(3) = [1, 0, 0]^T$$

b) backpropagation.

$$y = [0\ 0\ 1]^T$$

$$\delta 3 = a(3) - y$$
$$= [1, 0, 0] - [0, 0, 1]$$
$$= [1, 0, -1]^T$$

2) hidden layer

$$\delta 2 = w(3)^T \delta 3 \circ a(2)$$
$$\delta 2 = [0, 1]^T \circ [1, 1]^T$$
$$= [0, 1]^T$$

step 3 hidden layer 1 error delta(1)

$$\Delta 1 = [3,4]^T \circ [1,0]^T$$

$$= [3,0]^T$$

step 4 : eta = 0.5

layer 3 :

$$w(3)_{new} = [[-1,-5.5],[-3,-3],[5,8.5]].$$

$$b(3)_{new} = [0,-4,-2] - 0.5 * [1,0,-1]$$

$$= [-0.5,-4,-1.5]$$

layer 2 :

$$w(2)_{new} = [[1,-2],[0.5,4]]$$

$$b(2)_{new} = [1,0] - 0.5 * [0,1]$$

$$= [1,-0.5]$$

layer 1

$w(1)_{new} = [[-.5, -.5, -.5],$

$[3, 4, -2]$

$b(1)_{new} = [1, -2] - 0.5[3, 0]$

$= [-.5, -2].$

problem 2: Backprop for multiclass.

$\dot{\underline{A}}_{ij} = \dot{a}_{ij}$'s

$\text{grad}_a C = [-y_1/a_1, -y_2/a_2$

$\cdots\cdots -y_m/a_m]^T$

$\dot{a}_{ij} = a_i(1-a_i)$ if $i=j$

and $-a_i * a_j$ if $i \neq j$

$$\Delta_i = \text{sum}_j \left(\frac{\partial a_j}{\partial s_i}\right) \times \frac{-y_i}{a_j}$$

$$= -y_i + a_i \times \text{sum}_j\, y_j$$

$\text{sum}_j\, y_j = 1$ ← one hot constraint

$\therefore$ $\text{delta}_i = -y_i + a_i \times 1$

$$= a_i - y_i$$

$\text{delta} = a - y$ QED